

LOGIC - HOMEWORK

- 1) Determine if the proposition $(P \text{ or } Q)$ is logically equivalent to the proposition $(\text{not } P) \Rightarrow Q$.
- 2) Determine if the proposition $[(P \Rightarrow Q) \Rightarrow R]$ is logically equivalent to the proposition $[(\text{not } P) \Rightarrow R] \text{ and } (Q \Rightarrow R)$.
- 3) Construct the truth table of each of the following propositions:
 - (a) $(P \text{ or } Q) \text{ and } [\text{not } (P \text{ and } Q)]$
 - (b) $(P \text{ and } Q) \Rightarrow (P \text{ or } Q)$
 - (c) $[\text{not } (P \Rightarrow Q)] \text{ and } [(\text{not } Q) \Rightarrow (\text{not } P)]$
 - (d) $[Q \text{ or } (\text{not } P)] \Rightarrow (\text{not } R)$
 - (e) $[P \text{ and } (Q \text{ or } R)] \Rightarrow [(P \text{ and } R) \text{ or } Q]$
 - (f) $[P \text{ and } (\text{not } R)] \text{ or } [\text{not } (P \Rightarrow Q)]$
 - (g) $[(\text{not } Q) \text{ and } R] \Rightarrow [(\text{not } R) \text{ or } P]$

Answer key:

1) The two propositions are logically equivalent.

2) The two propositions are logically equivalent.

3)

(a)

P	Q	(P or Q) and [not (P and Q)]
T	T	F
T	F	T
F	T	T
F	F	F

(b)

P	Q	(P and Q) $\Rightarrow$ (P or Q)
T	T	T
T	F	T
F	T	T
F	F	T

(c)

P	Q	[not (P $\Rightarrow$ Q)] and [(not Q) $\Rightarrow$ (not P)]
T	T	F
T	F	F
F	T	F
F	F	F

(d)

P	Q	R	[Q or (not P)] $\Rightarrow$ (not R)
T	T	T	F
T	T	F	T
T	F	T	T
T	F	F	T
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	T

(e)

P	Q	R	$[P \text{ and } (Q \text{ or } R)] \Rightarrow [(P \text{ and } R) \text{ or } Q]$
T	T	T	T
T	T	F	T
T	F	T	T
T	F	F	T
F	T	T	T
F	T	F	T
F	F	T	T
F	F	F	T

(f)

P	Q	R	$[P \text{ and } (\text{not } R)] \text{ or } [\text{not } (P \Rightarrow Q)]$
T	T	T	F
T	T	F	T
T	F	T	T
T	F	F	T
F	T	T	F
F	T	F	F
F	F	T	F
F	F	F	F

(g)

P	Q	R	$[(\text{not } Q) \text{ and } R] \Rightarrow [(\text{not } R) \text{ or } P]$
T	T	T	T
T	T	F	T
T	F	T	T
T	F	F	T
F	T	T	T
F	T	F	T
F	F	T	F
F	F	F	T